TEST CAMPAIGNS: DP STEEL SHEET

One material (one steel sheet), per strain rate. Source: Beerli et al. (2026), Int. J. Impact Eng. 215, 105761, sections 2.2, 4.1 and 4.2. Lab names of the catalogue (10/11/2026). RD = rolling direction.

Not sure: inferred from the paper or to choose. The last column gives the question (end of the document).

1. SLOW RATE (QUASI-STATIC)

UT: 0.001 /s · other specimens: 6.6e-6 m/s · 100 kN hydraulic testing machine

SPECIMEN	ORIENTATIONS TO RD (°)	NO. ORIENT.	REPEATS	NUMBER	QUESTION
Tension-type specimens (waterjet)					
UT40	0, 15, 30, 45, 60, 75, 90	7	3	21	
NT20	0, 45, 90	3	3	9	
NT06	0, 45, 90	3	3	9	
CHD48 (CH5)	0, 45, 90	3	3	9	Q1, Q6
MNAK_RECT_FULL (punch)	0, 45, 90	3	3	9	Q5
MNAK_RECT_HOLE (mini-Nakazima)	0, 45, 90	3	3	9	Q5
Subtotal				**66**	
Shear (wire EDM)					
SHMD	0, 45, 90	3	2	6	Q4
Subtotal				**6**	
TOTAL				**72**	

2. MEDIUM RATE (INTERMEDIATE)

UT: 1 /s · other specimens: 6.6e-3 m/s · Instron 8801 (100 kN)

SPECIMEN	ORIENTATIONS TO RD (°)	NO. ORIENT.	REPEATS	NUMBER	QUESTION
Tension-type specimens (waterjet)					
UT40	0	1	3	3	Q2
NT20	0	1	3	3	Q2
NT06	0	1	3	3	Q2
CHD48 (CH5)	0	1	3	3	Q1, Q2, Q6
Subtotal				**12**	
Shear (wire EDM)					
SHMD	0	1	2	2	Q4
Subtotal				**2**	
TOTAL				**14**	

3. HIGH RATE

UT: 100 /s · other specimens: about 1.4 m/s · split Hopkinson tensile bar

SPECIMEN	ORIENTATIONS TO RD (°)	NO. ORIENT.	REPEATS	NUMBER	QUESTION
Tension-type specimens (waterjet)					
UT19 (small UT)	0	1	3	3	Q2
NT20	0	1	3	3	Q2
NT06	0	1	3	3	Q2
CHD48 (CH5)	0	1	3	3	Q1, Q2, Q6
Subtotal				**12**	
Shear (wire EDM)					
SHMD	0	1	2	2	Q4
Subtotal				**2**	
TOTAL				**14**	

All rates: 72 + 14 + 14 = **100 specimens per material**.

QUESTIONS FOR THE SUPERVISORS

Q1 **MATERIAL.** Which steel grade and which sheet thickness is this campaign for? The grade sets the shear geometry (low, medium or high ductility). Grade and thickness together set the maximum force of the central-hole specimen, hence CH5 or CH8 (see Q6).

Q2 **REPEATS AT MEDIUM AND HIGH RATE.** 3 per tension-type specimen? The paper does not say; 3 is inferred from its 150 tests (5 materials × 2 rates × 5 specimen types × 3).

Q3 **ORIENTATIONS AT MEDIUM AND HIGH RATE.** Only 0° to the rolling direction, as in the paper?

Q4 **SHEAR.** Only SHMD (the paper uses the "medium ductility" geometry), or all three LD / MD / HD? Orientations 0, 45, 90° (paper) or also 135° (2025 plan)? (2 repeats: confirmed.)

Q5 **PUNCH AND MINI-NAKAZIMA.** Is MNAK_RECT_FULL the punch specimen (equi-biaxial) and MNAK_RECT_HOLE the dihedral mini-Nakazima (plane strain)? 0, 45, 90° × 3 for both? (The current 600 × 500 order has MNAK_RECT_FULL at 0° only.)

Q6 **CENTRAL HOLE.** Our catalogue central-hole specimen has a 4.8 mm hole. In the paper the hole is reamed to its final size after waterjet cutting, for a clean edge. (a) Is the 4.8 mm hole a pre-hole, to be reamed to 5 mm (CH5) after cutting? (b) For high-strength grades (DP1180, DP1470) the paper uses an 8 mm hole (CH8) to keep the force below the limit of the clamps. Does this material need CH8? (Then a new template is needed: it is not in the catalogue.)

Q7 **CLAMPING ENDS.** In the paper the ends of the specimens (the part held by the grips) have a different shape for each rate: one for the slow machine, one for the Instron, a fir-tree shape for the Hopkinson bar (Fig. 2f-h). All specimens of our catalogue have fir-tree ends. Do the grips of the slow and medium machines take these fir-tree specimens, or do the slow and medium specimens need other ends (other templates)?

Q1 first: it decides the central-hole type and the shear geometry.